

# The Human Digestive System

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Science 8 - Narra | Life Science Unit

Follow along in class - the guided notes sheet is in the Resources tab.

# Learning Targets

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- Trace the path of food through the digestive tract in order.
- Explain mechanical versus chemical digestion.
- Identify the role of each major organ of digestion.
- Describe healthy habits that protect the digestive system.

# What Is Digestion?

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- Digestion breaks food down into nutrients the body can absorb.
- Mechanical digestion physically breaks food into smaller pieces (chewing, churning).
- Chemical digestion uses enzymes to split large molecules into usable ones.
- The whole journey takes 24 to 72 hours from mouth to exit.

# Stop 1: The Mouth

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- Teeth grind food - mechanical digestion begins here.
- Saliva contains amylase, an enzyme that starts breaking down starch.
- The tongue shapes food into a bolus and pushes it toward the throat.

## Stop 2: Esophagus and Stomach

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- Peristalsis: rhythmic muscle contractions squeeze food downward.
- The stomach stores, mixes, and churns food for about 2-4 hours.
- Gastric juice (hydrochloric acid + pepsin) digests protein into chyme.

## Stop 3: The Small Intestine

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- Most chemical digestion AND nutrient absorption happen here.
- The liver sends bile to emulsify fats; the pancreas sends enzymes.
- Villi - millions of tiny folds - maximize the absorbing surface area.

## Stop 4: The Large Intestine

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- Absorbs water and minerals back into the bloodstream.
- Houses gut bacteria that produce some vitamins.
- Compacts the remaining waste into stool for elimination.

# Care for Your Digestive System

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- Eat fiber-rich food: fruits, vegetables, whole grains.
- Drink enough water throughout the day.
- Chew slowly and avoid skipping meals.
- Wash hands before eating to prevent infections.



# Quick Check

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- Where does chemical digestion of starch begin?
- Which organ produces bile, and where is bile stored?
- Why are villi important in the small intestine?